

ConvNet Size & Parameters

Simple · ANSWER KEY

One black-and-white image through a small ConvNet - the lecture example.

THE ONLY TWO FORMULAS (stride 1, no padding)

Conv2D: $\text{params} = ((M \times N \times L) + B) \times F$ $\text{new size} = \text{old size} - (M - 1)$
 MaxPool: $\text{params} = 0$ $\text{new size} = \text{old size} / \text{pool}$
 Flatten: $\text{params} = 0$ $\text{size} = \text{rows} \times \text{cols} \times \text{channels}$
 Dense: $\text{params} = (\text{inputs} + 1) \times \text{nodes}$ $\text{size} = \text{nodes}$

M, N = kernel size L = channels coming IN (inherited, never chosen)
 B = bias = 1 F = filters = feature maps coming OUT (you choose)

THE CODE (given)

```
model = models.Sequential()
model.add(layers.Conv2D(3, (5, 5), activation='relu',
                        input_shape=(28, 28, 1)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(5, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Flatten())
model.add(layers.Dense(256, activation='relu'))
model.add(layers.Dense(2, activation='softmax'))
```

Layer	Output shape	Params	The math
Conv2D: F=3, (5x5)	(24, 24, 3)	78	$28 - (5 - 1) = 24$; $((5 \times 5 \times 1) + 1) \times 3$
MaxPool 2x2	(12, 12, 3)	0	$24 / 2 = 12$; nothing learned
Conv2D: F=5, (3x3)	(10, 10, 5)	140	$12 - (3 - 1) = 10$; $((3 \times 3 \times 3) + 1) \times 5$ L=3 in!
MaxPool 2x2	(5, 5, 5)	0	$10 / 2 = 5$
Flatten	(125,)	0	$5 \times 5 \times 5 = 125$
Dense(256)	(256,)	32,256	$(125 + 1) \times 256$
Dense(2, softmax)	(2,)	514	$(256 + 1) \times 2$
TOTAL trainable parameters		32,988	